

Analysis of Factors Affecting Restaurant Inspection Score in New York City

1. Introduction, Background and Research Questions

1.1 Introduction and Background

The inspection score for restaurants are a critical indicator of the public health compliance and food safety in the urban areas. These score are particularly useful since they can help in preventing serious health problems and hygiene issues (Kovacs et al., 2020). In the New York City which is a hub of diverse culinary, the score of inspection work as an evaluation metric of adherence of the restaurants towards health regulations. The Department of Health and Mental Hygiene (DOHMH) has been conducting these regular inspections and assigning score to the restaurants based on their regulatory following or violations. A higher score in this context indicates a poor compliance of the regulations which can potentially mean that the restaurant can have health risks for patrons. The scope of this study is to analyze these inspection score from 2024 to find trends and factors influencing the compliance. Through examining the score across boroughs, the types of cuisine and time the report aims at finding actionable insights for public health initiatives. New York City's diversity provides with a unique opportunity for exploring variations in the compliance of regulations across its five boroughs along with a wide array of cuisine types ranging from local American to global Asian and Italian American flavors.

1.2 Research Questions

This study is majorly focused on the following research questions:

- i. **Borough and Cuisine Variations:** How do inspection score vary across the boroughs and variety of cuisine types?
- ii. **Temporal Trends:** Is there a temporal trend in restaurant inspection score in New York City during 2024?
- iii. **Factors Affecting Score:** What are the main factors associated with higher inspection score, indicating poorer health compliance?

2. Data Description

The dataset that is being utilized in this evaluation is the Department of Health and Mental Hygiene (DOHMH) in the New York City. Only the inspections that were conducted in the year of 2024 were considered for the context of this study (NYC Open Data, 2024). This data is sourced from the New York City Data platform. It contains detailed information about

the inspections conducted by the Department of Health and Mental Hygiene (DOHMH) in New York City on restaurants. The key variables are mentioned below:

- **SCORE:** The total inspection score assigned where a higher values indicating poorer compliance and with health standards.
- **BORO:** This variable is the borough in which the restaurant is located.
- **CUISINE DESCRIPTION:** This shows the type of cuisine that is offered by the restaurant like Italian, Chinese, and American.
- **INSPECTION DATE:** It is the date of the inspection that can be used in analyzing the temporal trends.
- **VIOLATION CODE and VIOLATION DESCRIPTION:** These are the detailed codes and description of the specific type of violations that are recorded during the inspections.
- **CRITICAL FLAG:** This is a binary indicator that shows whether the violation is classified as a critical violation or not.
- **INSPECTION TYPE:** The type of the inspection conducted like initial or follow-ups.

3. Data Cleaning

1. **Outlier processing:** Outliers were identified using the interquartile range (IQR) method across key variables. Records with operational significance, such as high scores indicative of serious sanitation issues, were retained. Obvious errors were corrected, and robust modeling methods were employed to minimize the impact of extreme values.
2. **Feature engineering:** Feature engineering played a critical role in enhancing the dataset's analytical potential. Inspection dates were transformed into month and season categories, enabling the identification of temporal trends. Administrative districts and restaurant types were encoded, and new composite metrics were created. These included the region score difference (BORO_SCORE_DIFF), differences in meal type scores, and trends in average score variations, providing additional context for understanding compliance patterns.
3. **Data Standardization:** To ensure consistency and comparability, numerical features were normalized, and categorical variables were encoded while maintaining interpretability. These standardization efforts preserved the original data structure while improving its usability for modeling and analysis.

4. Exploratory Data Analysis

This section of the report contains information about the exploratory data analysis (EDA) conducted on the dataset to find patterns and trends along with any significant differences in restaurant inspection score across various boroughs, cuisines and time frames. The exploration of the data was majorly focused on the three research questions.

4.1 Borough Analysis

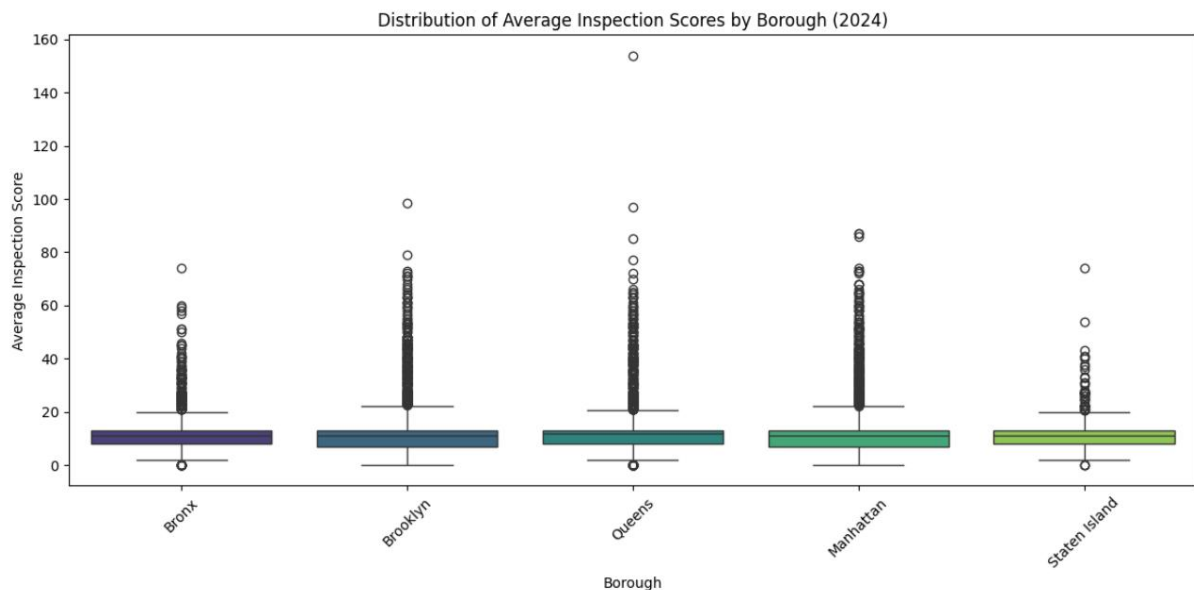


Figure 1: Distribution of Average Inspection Scores by Borough (2024)

As shown in Figure 1, the performance of restaurant inspection scores across New York City's administrative districts reveals several notable patterns. The median scores are relatively similar across boroughs, suggesting a level of uniformity in maintaining basic health standards citywide. Staten Island stands out with the least volatile scores, indicating that enforcement standards in this borough are likely the most consistent. However, significant outliers, with inspection scores exceeding 60, are present in all boroughs, highlighting severe sanitation issues in individual establishments. Brooklyn and Queens exhibit the widest distribution of outliers, likely due to their large number and diverse range of restaurants, which may pose unique challenges in maintaining consistent compliance. These findings underscore the need for tailored enforcement strategies to address borough-specific dynamics.

For exploring the variations in the compliance based on the boroughs the average inspection score were calculated for each of the borough. It was noted that the Queens has the highest

average score which indicates that it has the poorest compliance. The Staten Island has the lowest average score while the Manhattan, Brooklyn and Bronx are falling in the moderate range.

4.2 Cuisine Analysis

Figure 2 highlights the performance of various restaurant types based on their average inspection scores in 2024. Iranian cuisine had the highest average score of approximately 29 but also exhibited the largest standard deviation, suggesting significant variability in quality and compliance across establishments. Filipino cuisine followed with an average score of about 24 points. In contrast, Indian, Chinese/Cuban, and Eastern European cuisines demonstrated smaller standard deviations, indicating more consistent adherence to health standards. These differences in average scores and variability may reflect the diverse cooking techniques and ingredient handling requirements associated with different cuisines, which can influence the likelihood of compliance with health and safety regulations.

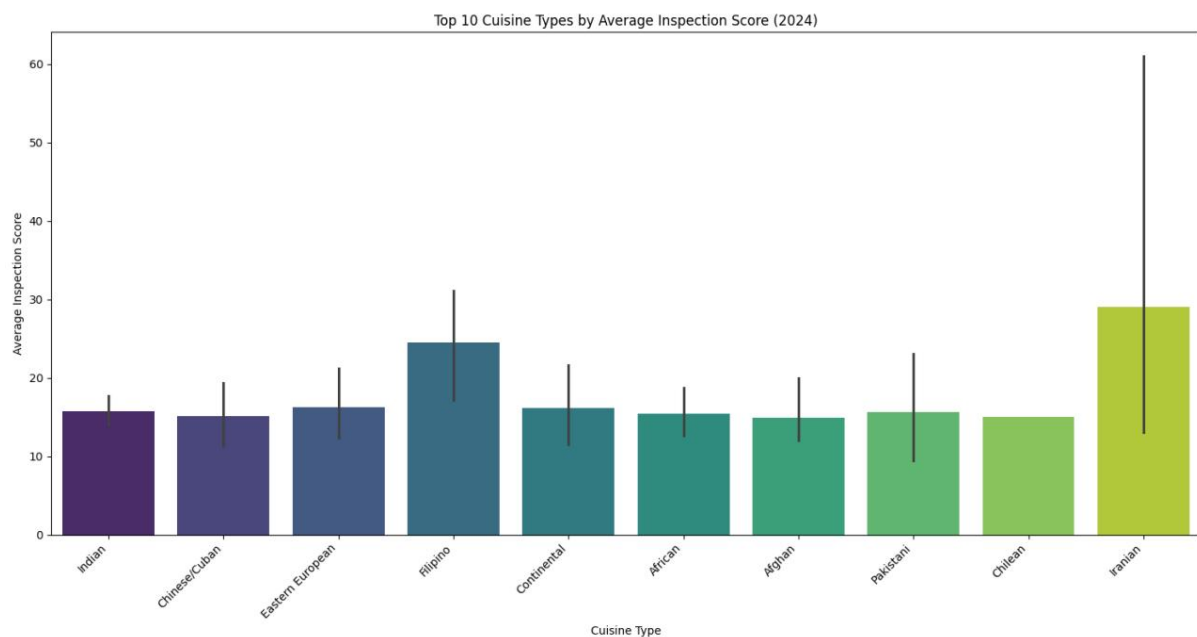


Figure 2: Top 10 Cuisine Types by Average Inspection Score (2024)

4.3 Temporal Trends

The monthly trends in the inspection score show a seasonal pattern as observed in the Figure 3. It can be noted that the score saw a significant dip in the in the middle of the year, in the month of June, thereby showing a better compliance and less number of violations across the

boroughs. It can be deciphered that June is a notable month for compliance improvement for the year 2024, and this may be influenced by seasonal factors (e.g., temperature, humidity)

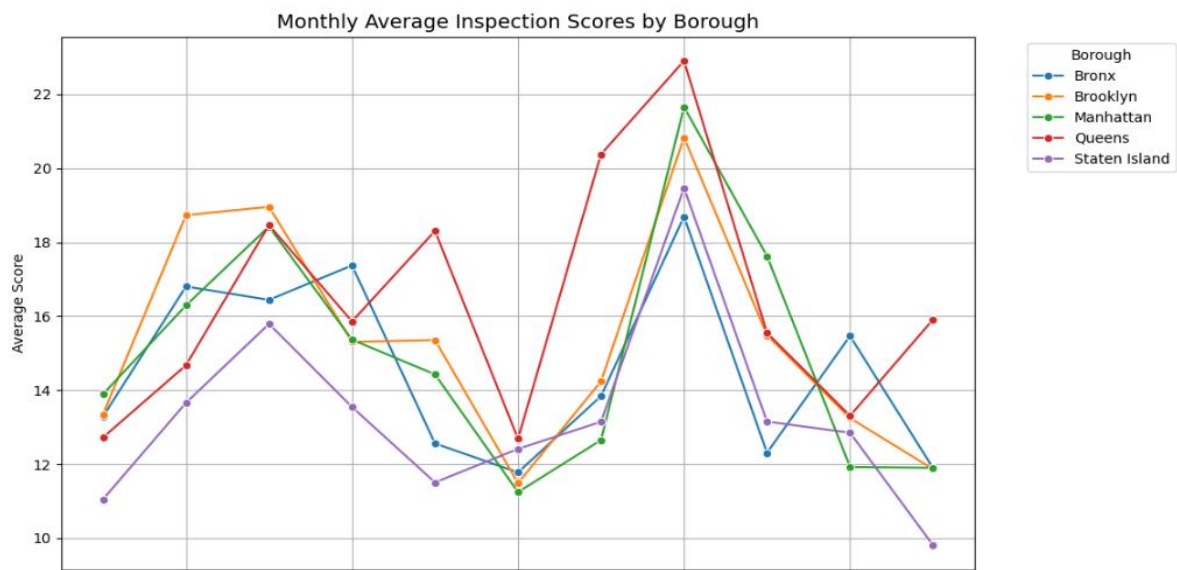


Figure 3: Monthly Average Inspection Score by Borough

5. Prediction Analysis

For identifying the key factors that are most influential in the restaurant inspection score the regression model was used. This section of the report shows the model setup for regression, the results obtained and the insights derived from those results.

5.1 Data Splitting

- The dataset was split into training and testing sets using stratified sampling. The stratification was based on quantiles of the target variable (y) to ensure a representative distribution across both sets.
- The test set comprised 20% of the data, and a fixed random seed (random_state=42) was used for reproducibility.

5.2 Model Performance Summary

The Table 1 shows the performance metrics for each of the model being applied on the dataset.

The comparison of the models are also shown in the Figure 4 where it can be noted that the Random Forest model outperforms the linear model based on its highest Test R^2 value of 0.8487 and the lowest test RMSE value of 3.5402. Specifically, it exhibits a 15% improvement in prediction accuracy over linear models and a 24% reduction in prediction error. Additionally, the model shows robust performance across cross-validation folds, with no significant overfitting evidenced by a minimal train-test gap of 0.022.

Table 1: Regression Results

Model	Train R^2	Test R^2	Train RMSE	Test RMSE	Train MAE	Test MAE	CV R^2 (mean $\pm$ std)
Linear Regression	0.7395	0.7364	4.4542	4.6731	2.1906	2.2930	0.7397 ± 0.0470
Ridge Regression	0.7395	0.7363	4.4543	4.6735	2.1885	2.2918	0.7397 ± 0.0470
Lasso Regression	0.7393	0.7361	4.4559	4.6752	2.1829	2.2876	0.7396 ± 0.0473
Random Forest	0.8705	0.8487	3.1401	3.5402	0.8379	1.0109	0.8385 ± 0.0461

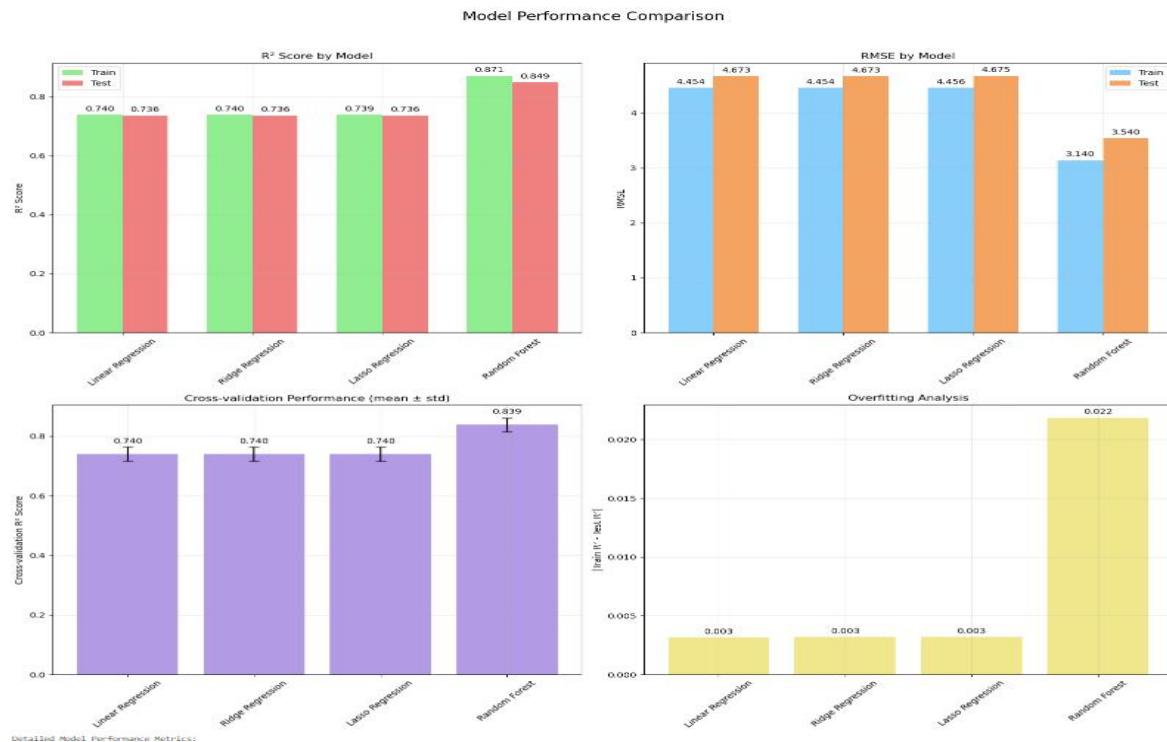


Figure 4: Model Performance Comparison Chart

5.3 Residual Analysis

Analysing the residuals for validating the assumptions made by the model is crucial. This also helps in identifying any patterns or biases in the prediction. Figure 5 shows the actual vs. predicted scatter plot for the Random Forest model which shows that it closely aligns between the actual and predicted values while having minimal level of deviations. The residual distribution in the Figure 5 shows that they are tightly distributed around the zero, indicating consistency in the prediction results. Further, it shows the Q-Q plot where the Random Forest residuals are seen to be aligning well with the theoretical quantile.

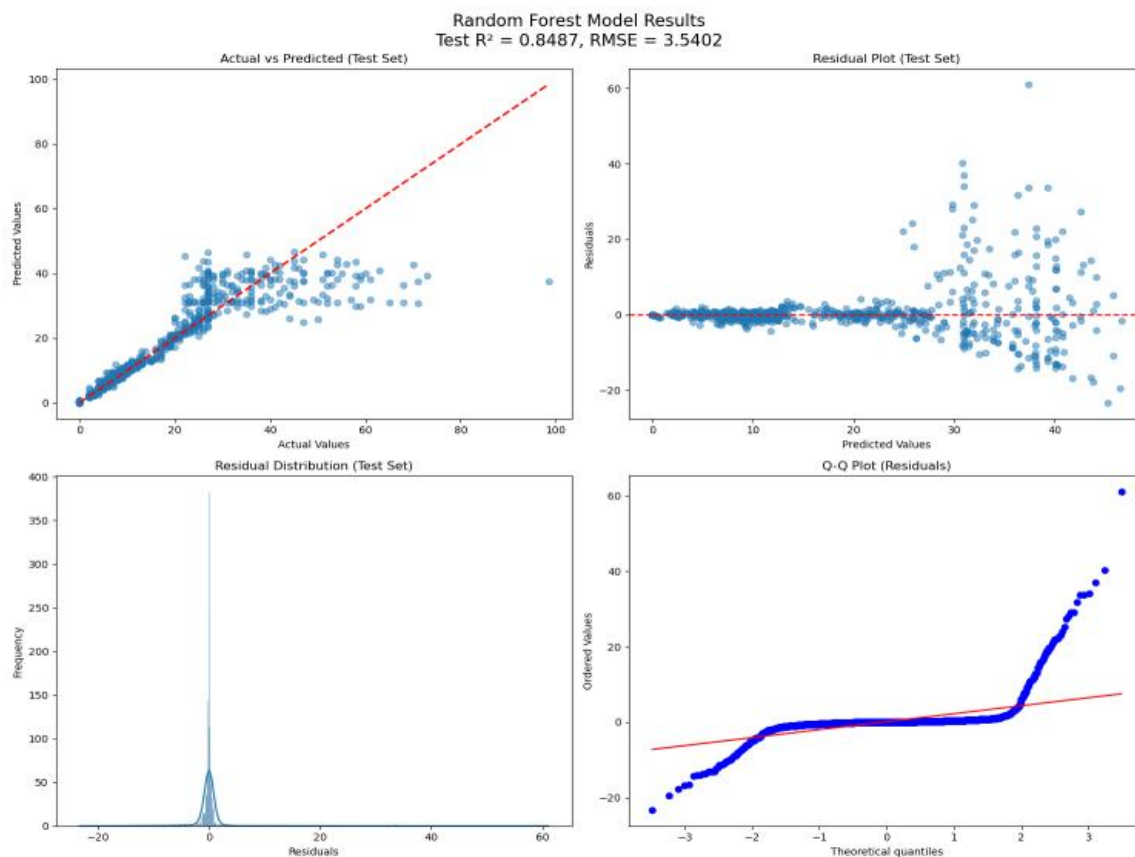
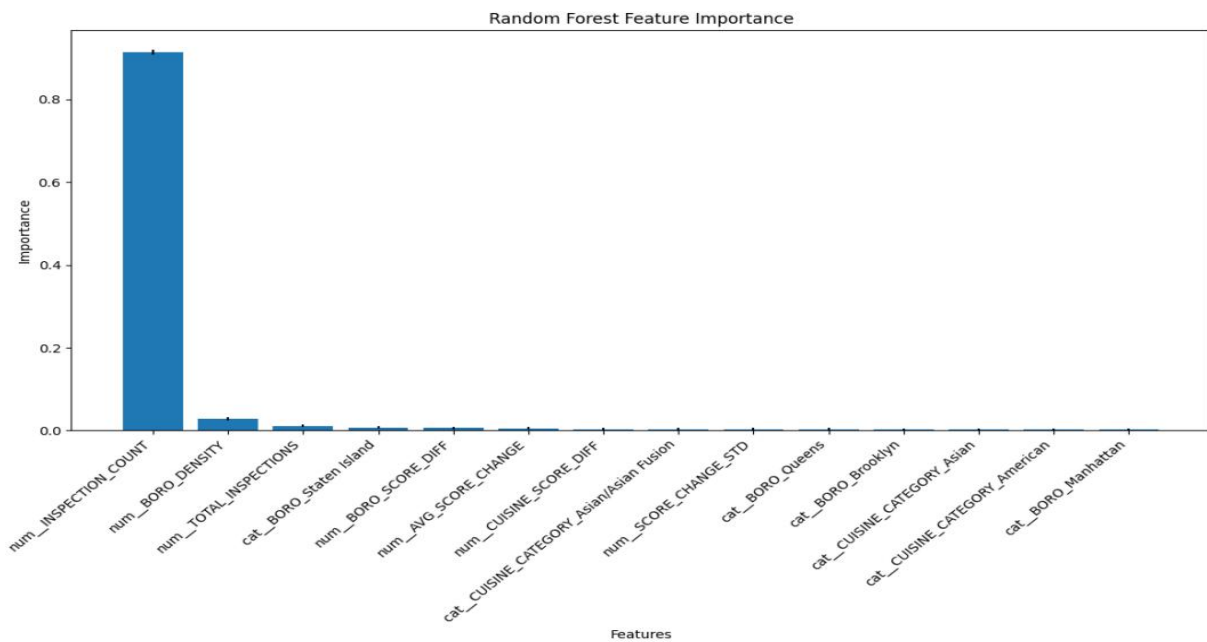


Figure 5: Residual Analysis plots

5.4 Key Influencing Factors



The analysis identified the most critical predictors of restaurant inspection scores, with the frequency of inspections (INSPECTION_COUNT) emerging as the dominant factor. This variable accounted for 91.44% of the predictive importance, far surpassing all other features. The strong correlation between inspection frequency and food safety performance suggests that restaurants subjected to frequent inspections often have a history of non-compliance or significant health code violations.

In addition to inspection frequency, secondary factors played a much smaller role in influencing inspection scores:

- **Regional Population Density** (BORO_DENSITY): Contributed 2.92% to the predictive importance, indicating that environmental and demographic factors of the restaurant's location can influence compliance trends.
- **Total Number of Inspections** (TOTAL_INSPECTIONS): Accounted for 1.13%, reflecting the overall inspection activity in the area.
- **Geographic Location on Staten Island**: Represented 0.77%, suggesting that specific regional characteristics or regulatory practices may slightly impact performance.
- **Regional Scoring Differences** (BORO_SCORE_DIFF): Contributed 0.63%, highlighting minor variations in scoring trends across boroughs.

Frequency of inspections may reflect a restaurant's historical compliance record, and restaurants that receive frequent inspections may have a history of past violations. Regional population density, the second most important factor, suggests that environmental characteristics of the area where the restaurant is located have an impact on food safety. Geographic location (e.g., whether or not it is on Staten Island) also has a slight impact on restaurant performance.

6. Discussion

The findings from the analysis shows valuable insights about the factors influencing restaurant's inspection score in the New York City. The analysis had the following contributions and implications:

6.1 Contribution

Predictive Capability: A key outcome of this study was the development of a highly accurate predictive model with an R^2 value exceeding 0.84. This model can serve as a foundation for early warning systems to identify restaurants at higher risk of non-compliance. By flagging potential problem establishments, the model enables proactive intervention to improve food safety standards and protect public health.

Policy Implications: The findings highlight the need for greater harmonization of enforcement standards across regions to ensure consistency in compliance monitoring. Additionally, specific types of restaurants may require tailored regulatory approaches due to their unique characteristics and risks. Seasonal factors, which were shown to impact inspection outcomes, should also be considered when scheduling and conducting inspections, as they may influence compliance trends.

6.2 Suggestion for Improvement

Regulatory Priorities

- Increased regulation of restaurants with a history of inconsistent performance
- Focus on restaurants in high-density areas
- Develop specialized regulatory strategies for specific restaurant categories

Resource Allocation

- Adjust inspection frequency based on regional score differences
- Increase inspection frequency for restaurants with fluctuating performance

- Provide targeted guidance for specific types of restaurants

Policy Optimization

- Establish a dynamic scoring mechanism
- Improve regional differentiation of regulatory strategies
- Develop meal-specific assessment criteria

6.2 Research Limitations

Data Limitations: The dataset lacked specific violation details and did not include important operational variables such as restaurant size, hours of operation, or staffing levels. These missing factors could potentially influence compliance outcomes. Furthermore, the methods used for outlier handling, while robust, may have impacted the accuracy and generalizability of the model.

Methodological Limitations: The analysis did not account for spatial autocorrelation, which could affect the independence of observations based on geographic location. Additionally, while temporal trends were explored, a deeper time series analysis could provide more granular insights into seasonal and long-term compliance patterns. Lastly, the study focused on correlations rather than causality, leaving the underlying drivers of compliance trends to be further investigated.

6.3 Conclusion

By analyzing restaurant sanitation inspection data in New York City, this study found that geographic location, restaurant type, and seasonal factors all significantly affect inspection results. The random forest model demonstrated good predictive ability and can be used to assist regulatory decision-making. The results of the study have important reference value for optimizing the inspection system and improving regulatory efficiency. It is recommended to further improve the data collection and deepen the analysis methods to provide stronger support for ensuring food safety.

References

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